Auralization of Impulse Responses Modeled on the Basis of Ray-Tracing Results*

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After a short introduction to auralization the most common methods of acoustic room simulation—ray tracing and the image source model—are discussed briefly. It is shown how the information contained in an energy histogram obtained by ray tracing can be converted into an "equivalent impulse response" which, after binaural processing, is ready for auralization sufficient for the auralization process. Furthermore the steps are discussed which are needed to transform this information into a binaural impulse response ready for a convolution with music signals.

0 INTRODUCTION

The term "auralization" denotes a new procedure which enables us to create aural impressions from computer-simulated room responses. The simulation itself can be based on acoustic and geometric data of existing as well as nonexisting rooms. Thus auralization can be used to process music in such a way that it sounds as if performed in a hall that is still under design, or even in a completely hypothetical hall which no architect would ever take into consideration. Furthermore auralization is becoming an important tool in room acoustic research since it allows us to study the influence of size, shape, materials, arrangement of audience seats, and many other factors on the acoustics of a hall.

The idea of auralization is quite old. As early as in 1934 Spandöck [1] proposed to replay frequency-transformed music signals in a scale model, pick them up again in the model, and present them by earphone to a listener after retransformation into the original frequency range. A similar procedure using computer models was proposed by Schroeder and coworkers [2] in 1961 and was first applied to simple rectangular enclosures by Allen and Berkley [3]. The first binaural auralization based on rectangular rooms is due to Pösselt [4]. Since then several authors have reported on auralization schemes and the results they obtained with them. (See, for instance, [5]–[10].)

Basically auralization consists in the construction or simulation of a digital filter which reproduces both the transmission of sound signals in the room considered and their binaural reception by the listener. The latter requirement is accounted for by including the listener's personal head transfer functions into the filter characteristics.

The present paper concentrates on the way in which the room response is simulated and how the results of this simulation are transformed into the characteristics of an auralization filter.

1 BASIC METHODS OF ROOM SIMULATION

To simulate the physical transmission of sound signals in an enclosure, basically two different procedures are currently used, ray tracing and the image source method.

In a ray-tracing procedure as first applied to concert hall acoustics by Krokstad, Strøm, and Sørsdal [11], a sound source is imagined to release numerous sound particles into all directions at a certain moment. The path of each particle, including its reflections from specularly or diffusely reflecting walls, is followed up with the computer, and whenever it arrives at a previously assigned "counter" (a plane or spherical area, for instance), its energy, arrival time, direction, and so on, are registered. The result of this process can be presented as a histogram (Fig. 1) showing the temporal distribution of the energy received and can be regarded

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as an approximation to the energetic impulse response of the enclosure, with the degree of approximation depending on the number of particles and the time resolution achieved.

The image source method is based on the old idea that a sound ray which is reflected from a plane wall can be thought of as originating from an image source which is the mirror image of the original sound source formed by the wall. If there is more than one wall, in particular if there are several or many plane walls making up an enclosure, the mirroring process has to be applied to image sources already constructed, leading to second-order, third-order, etc. images. The final basically infinite pattern of the image sources represents the original room, so to speak. The tedious process of tracing individual ray paths is replaced by just adding the contributions of all image sources, taking into account the $1/r^2$ law, wall absorption, and air attenuation. If the assumed sound signal is a Dirac impulse, the result of this process is the correct impulse response of the enclosure. An example, calculated for frequency-independent reflectivities of all walls, is shown in Fig. 2.

To obtain realistic impulse responses, however, the frequency dependence of wall reflectivities (and air attenuation) must be accounted for in some way, for instance, by convolving each spike in Fig. 2 with the "reflection responses" $r_n(t)$ of all the walls involved in the formation of that particular component. Here $r_n(t)$ is the Fourier transform of a wall's complex reflection factor $R_n(x)$.

At first glance ray tracing results as shown in Fig. 1 are less suitable as a basis for the digital filter we need for auralization since they represent not impulse responses but just temporal energy distributions. Furthermore this method is unable to model strong and isolated components caused by specular reflections from smooth and extended walls. But the image source model also has its shortcomings. For one, it is restricted to specular reflections. Furthermore the number of significant image sources grows exponentially with the desired length of the impulse response, and only a small fraction of all image sources is "visible" from a given receiving point [12]. (Nowadays various schemes are available to eliminate the invisible images [13]–[16].) Finally, when applied to the total room impulse response, the image source method yields much more information on sound transmission than is significant from a psychoacoustical point of view.

2 MODELING THE IMPULSE RESPONSE

As discussed in the preceding section, both methods of simulation—ray tracing and the image source method—have their specific merits and shortcomings. Therefore it seems reasonable to combine both in such a way that an auralization filter based on the result of the simulation would create exactly the same subjective impression as the original response, that is, music passed through this filter sounds the same way as it would sound in the hall considered.

To do this, one should keep in mind that two portions of a room's impulse response with different subjective effects have to be distinguished (see Fig. 3). The first consists of the direct sound and the earliest reflections. Its temporal and directional structure is of crucial importance for the sensation of loudness, clarity, spaciousness, and so on [17]. It is also responsible for the variations in listening conditions at different parts of a hall. Quite frequently this part consists mainly of strong peaks due to specular reflections from extended walls or wall portions, but even then it contains some diffuse background caused by sound scattered from wall irregularities, edges, seats, and so on.

The remaining part, the "late portion" of the impulse response, consists of numerous overlapping reflections. This is what is usually known as reverberation. It is also important for the hearing impression, but rather on account of its gross temporal and spectral characteristics than of its fine structure. Furthermore it does not contain much information on the directional structure of the sound field. Generally the directional distribution of a sound field in a room is quite irregular immediately after the direct sound has arrived, but with increasing time delay of the incoming reflections it

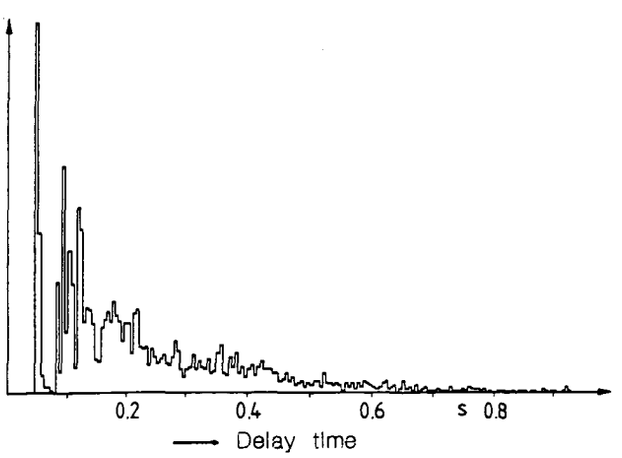

Fig. 1. Typical ray-tracing histogram (linear energy scale).

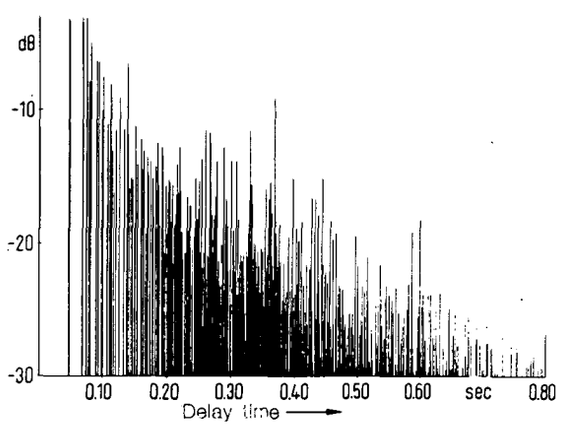

Fig. 2. Energetic impulse response of a room, simulated using image source method for frequency-independent wall reflectivity.

becomes increasingly uniform due to the randomizing effect of the enclosure. In a hall of some size the sound field can be regarded as diffuse after 100–150 ms [18], [19]. Thus the characteristic time t_c which separates the early and late parts of the impulse response should be in this range.

Several authors have developed simulation schemes which combine ray tracing and the image source model in some way. One of the most recent ones is due to Heinz [10]. Here both simulation processes are operated in parallel, so to speak, the ray-tracing process being fed by nonspecularly reflected sound portions. (Other authors split up the total impulse response into three rather than two parts [9].)

There is one problem common to both ray tracing and the application of image sources. As is well known, sound waves propagating at grazing directions over audience are subject to considerable frequency-dependent excess attenuation. Although this effect strongly influences the strength of the direct sound arriving at a particular listener's ears, as well as of sidewall reflections, it is neglected in many published simulation schemes since only sparse data on it are available. Quite recently Mommertz has carried out measurements on this type of attenuation in several occupied halls using time-efficient maximum-length techniques [20]. His results seem to indicate that within a certain frequency range the strength of the sound signal decreases exponentially with increasing distance (which is not self-explaining), with an attenuation constant depending on the grazing angle.

Since ray-tracing results do not represent impulse responses but temporal energy distributions, they have to be converted into an "equivalent impulse response" with the same temporal and spectral composition, but with a completely new "microscopic" structure. This can be done in the following way [19].

The entire frequency range is subdivided into a number of frequency bands, say 10 octave bands, for instance. For each of them either a ray-tracing simulation or the combined process as described is carried out, with the proper wall absorption coefficient and air attenuation, while the geometric data (room shape, location of sound source and receiver, and so on) remain unchanged. A time resolution of 5 or 10 ms seems sufficient. The results can be represented by numbers E_{ik} (Fig. 4), with i and k denoting the number of the time interval and of the frequency band, respectively. Thus the set of values with constant i represents a short-time energy spectrum of reverberation in the time interval Δt_i. From this result a continuous function of frequency and time $E(f, \tau)$ is obtained by a proper smoothing process, which eliminates the statistical errors inherent in the ray-tracing procedure.

The next step is to find a time function $g_i(t)$ with practically the same energy spectrum as the simulated sound decay. It is obtained by taking the square root and Fourier transformation of $E(f, \tau)$. The latter step requires knowledge of the phase spectrum, strictly speaking, which is not contained in $E(f, \tau)$. Now a sequence of Poisson-distributed points t_n on the time axis is selected by generating exponentially distributed random numbers. Then the equivalent response in the time interval Δt_i is obtained by

$$g(t) = \sum_n g_i(t - t_n) \ . \tag{1}$$

The average density of t_n on the time axis should be at least 5000 per second.

Here a remark on the phase spectrum may be in order. It could be formed from $E(f, \tau)$, assuming the transmitting system to be of the minimum-phase type. However, several simulations performed by Becker and Mommertz [21] followed by careful listening tests have shown that the phase spectrum of the simulated auralization filter is subjectively insignificant—a consequence of the random superposition of the elements $g_i(t)$ according to Eq. (1). Thus the phase spectrum used for the above-mentioned Fourier transformation is noncritical and can be chosen in any reasonable way that is convenient from the computational point of view.

3 BINAURAL PROCESSING

A prerequisite for the creation of realistic hearing impressions by any auralization filter is the simulation of not only the sound transmission in the room itself,

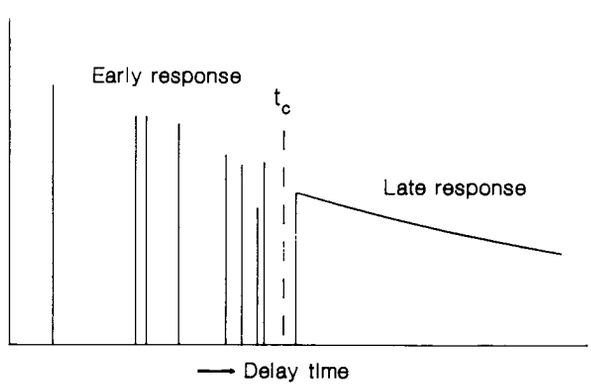

Fig. 3. Early and late portions of schematic room impulse response.

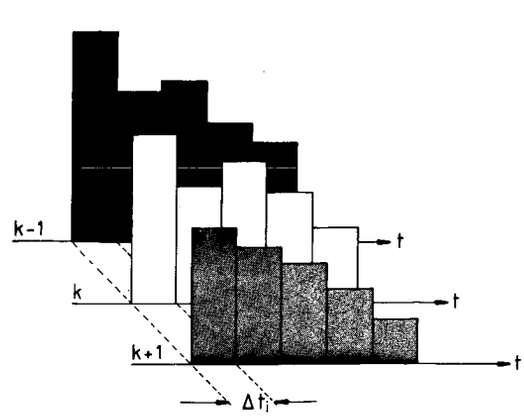

Fig. 4. Ray-tracing histograms for different frequency ranges. i—time interval; k—frequency band.

but also the diffraction of the sound waves around the listener's head. This effect depends strongly on the direction of sound incidence and is responsible for our spatial hearing. It is usually accounted for by including the so-called head transfer functions into the auralization filter. The result is a digital filter with two output terminals, one for each ear.

In this context two general questions arise. Strictly speaking, each contribution to the impulse response ought to be multiplied by the head transfer functions valid for its particular direction of incidence, which is impossible. Regarding the limited directional resolution of our hearing, it is sufficient to "discretize" directions, that is, to subdivide the directional space into finite directional groups and to attribute two representative head transfer functions to each of them. How many groups are necessary (and significant), and which are their locations and sizes? The answer to this question depends on the kind of component being considered. According to recent investigations [22] the direction of the direct sound has to be classified into one of 30 or 40 directional groups, whereas for the strong and early reflections five properly selected and overlapping groups are sufficient. The same holds for the "reverberant tail," whose components $g_i(t - t_n)$ [see Eq. (1)] are randomly attributed to one of five directional classes. The second question concerns the "individuality" of the head transfer functions, which show considerable variations for different persons. Fortunately, according to Moeller [23], these variations are largely due to anatomical differences of the ear channel and hence are not influenced by the diffraction process. Hence they can be eliminated by a filter, which must be adjusted individually, whereas the direction-dependent part of the head transfer function may be an averaged version. This facilitates binaural processing of the impulse responses considerably, and hence the whole process of auralization.

4 RESULTS AND LISTENING TESTS

Auralization filters designed after the scheme described have been programmed in various modifications by Heinz and coworkers (see, for example [10]). The simulated enclosure is a multipurpose hall having a volume of about 15 000 m^3 which is mainly used for concerts. Thus it is possible to compare music samples processed with an auralization filter based on that hall with those obtained by convolving the samples with the real impulse response as recorded binaurally for the same situation in the real hall.

But the most crucial way of validating the simulation is to compare not music or speech samples but the impulse responses themselves. (This corresponds to the old-fashioned hand-clapping test, which is still popular in a rough diagnosis of room acoustics.) If this comparison is successful, one can be sure that no problems will arise with music being processed by the auralization filter.

The performance and fidelity of auralization are improved greatly by replacing the usual earphone presentation of the processed sound with loudspeaker reproduction. In this way the well-known effect of "in-head" localization is avoided and a more natural listening impression achieved. Of course, the crosstalk between both channels caused by sound diffraction around the listener's head has to be compensated for in the manner carried out first by Atal and Schroeder [24]. Again, this method which has been investigated thoroughly and was improved by Schmitz [25], requires the knowledge of head transfer functions and can easily be incorporated into an auralization filter.

Listening tests carried out with realized versions of the auralization filter have shown that quite realistic hearing impressions can be conveyed by carefully designed auralization filters. Special care is needed for joining the early and late parts of the simulated impulse response, and the way the short-time spectra obtained by raytracing are smoothed is quite critical.

5 CONCLUSION

The modeling scheme described in this paper has considerable advantages over the pure image source method. In the first place, it requires less computing time than the latter. Furthermore it is capable of accounting for the numerous nonspecular components occurring in all realistic situations. This can be done by assigning to each wall the fraction of incident sound energy that will undergo diffuse reflection. It has turned out that no knowledge of the complex wall reflection factors is needed, which are difficult to collect. The same holds for the loss processes responsible for sound absorption in the air. Instead it is sufficient to base the design of an auralization filter on the absorption coefficients of the walls and on published air attenuation constants. Nevertheless, several details of the procedure need closer investigation, and further improvements and simplifications are certainly possible. But even in the present state of the art this method seems to meet expectations.

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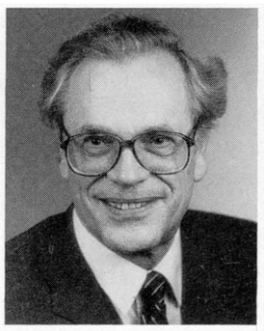

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